SA 220 AQUACULTURE

❖ FEW ANNOUNCEMENTS

- Classes will be commencing on the stipulated time and for the stated duration
- You are encouraged to attend classes and write your own notes

SA 220 AQUACULTURE

❖ **Course Materials:** You will be provided with the following materials.

➢ Lecture notes after the material has been delivered.

➢ List of recommended textbooks and relevant articles.

❑ Not compulsory for you to acquire or indeed read, But will help to supplement the course material.

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- **Assessment**

- There shall be a test in week two of learning (Agree on the date and time)

- We shall have an assignment (Individual)

SA 220 AQUACULTURE

- Fisheries Officer (MFL)- Ndola
- MSc Sustainable Aquaculture – University of Stirling, UK.
- MSc Aquaculture and Fisheries Science – LUANAR, Malawi.
- BSc Fisheries and Aquaculture –CBU, Zambia.

SA 220 AQUACULTURE



Course aim:

To provide the basic background information on Aquaculture with special reference to fish farming and also to link capture fisheries and aquaculture as contributing sectors to food fish as a cheap source of protein in human diets.

SA 220 AQUACULTURE



Course objectives:

- To know the different key terms used in aquaculture
- To appreciate the importance of aquaculture to society and SDGs
- To know and identify the various fish culture system
- To know the species of economic importance and factors considered when choosing them.

SA 220 AQUACULTURE

Course objectives:



- To have a basic knowledge of hatchery and pond management
- To explain the basics of fish feed formulation
- To know the food and feeding habits of different fish
- To understand the basics of artificial propagation of fish.

KEY TERMS

1. AQUACULTURE

- Aquaculture is the farming of aquatic organisms, including fish, mollusks, crustaceans and aquatic plants in a natural, semi-controlled or fully controlled manner

FAO (1990) https://www.fao.org/3/t0697e/t0697e02.htm

KEY TERMS

2. CRUSTACEANS



- Crustaceans are a very diverse group of invertebrate animals which includes active animals such as the crabs, lobsters, shrimp, krill, copepods, amphipods, and more sessile creatures like barnacles.

KEY TERMS

3. MOLLUSCS

Molluscs are soft bodied animals, often with hard shells for protection





KEY TERMS

4. FISH SEED OR FINGERLING

- A fish ‘seed’ is a juvenile fish to be used for fish farming



5. FISH BREEDING

- entails all the various techniques or methods used in fish seed production

KEY TERMS

6. BROODSTOCK/BROODER

- Brood stock, or brood fish are a group of mature individuals used in aquaculture for breeding purposes or brood fish, are the parent fish from which fry and fingerlings are produced

KEY TERMS

7. FISH HATCHERY

- A fish hatchery is a place where large numbers of fish eggs are artificially fertilized and fry are hatched in an enclosed environment.

KEY TERMS

8. HYPOPHYSATION

- Hypophysation is artificial or induced breeding, a technique that uses pituitary gland extractinjection



Plate 4. Ovatide: A synthetic hormone that can also be used.

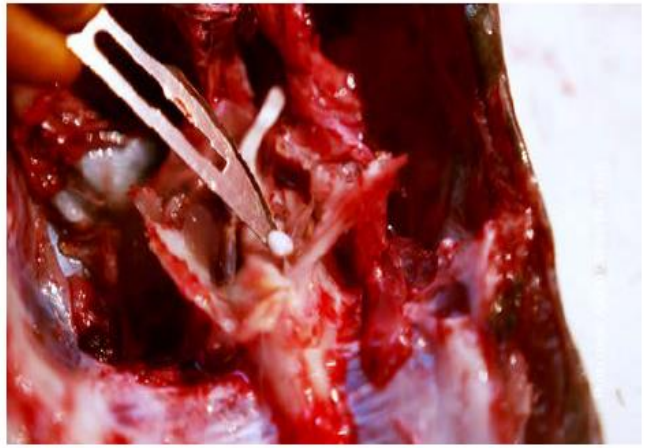


Plate 3. Pituitary gland being extracted.



Plate 5. Injecting a female broodstock above lateral line.

KEY TERMS

9. INGREDIENT

- A component part or constituent of any mixture making up a feed.

10. NUTRIENT

- The components in a diet that an organism utilizes to survive and grow.

KEY TERMS

11. FEED CONVERSION RATIO (FCR)

- This is the measures the efficiency with which an animal converts feed into its body mass.

Formulae:

FCR=Amount of Feed given / Amount of weigh gained by the animal

KEY TERMS

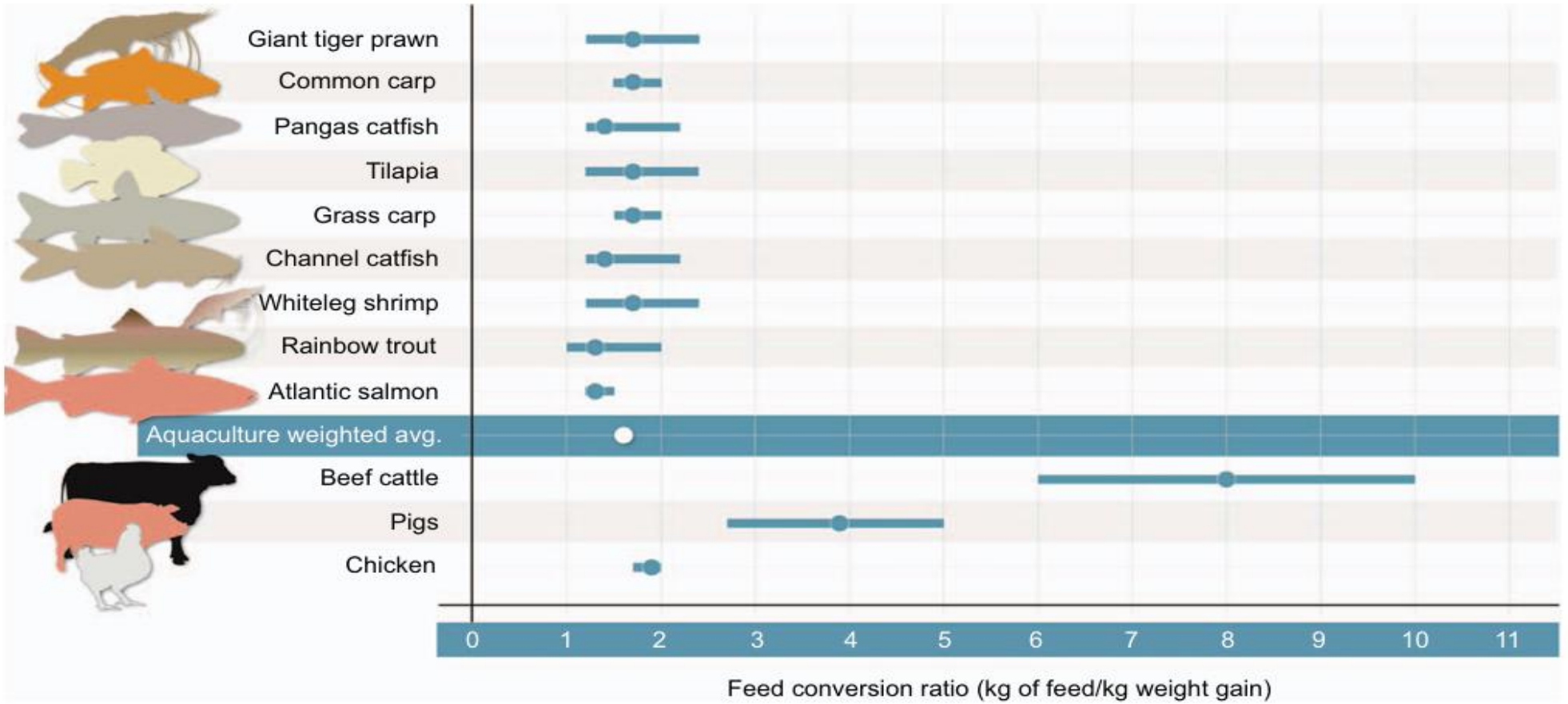


Figure 1. Feed conversion ratios for selected aquatic and terrestrial farmed animal species. Dots represent means and bars indicate range. Lower values signify higher efficiency. Sources: Tacon and Metian (2008) [12], Smil (2013) [13], Shike (2013) [14], Zuidhof *et al* (2014) [15], and Rabobank Research (2015) [16].

KEY TERMS

12. Natural fish food organisms - plankton, insects and other aquatic organisms that fish eat.

13. Phytoplankton - the plant component of plankton.

14. Plankton - the various, mostly microscopic, aquatic organisms (plants and animals) that serve as food for larger aquatic animals.

KEY TERMS

15. Pond dike - the wall of a pond which is constructed to hold in the water.

16. Predatory/carnivorous fish - a fish species that eats other fish.

17. Supplemental/incomplete feed - a feed that does not contain all the vitamins and nutrients essential for growth, and which is produced outside of the pond.

KEY TERMS

18. Omnivores- These are species that feed mainly on both plant and animal materials. Most Catfish species are mainly omnivores with diets ranging from algae, vegetable matter and insects.

19. Herbivores- These are the fishes that feed predominantly on higher plant materials including vegetable matter.

20. Monoculture - This is the culture of only one species of fish.

KEY TERMS

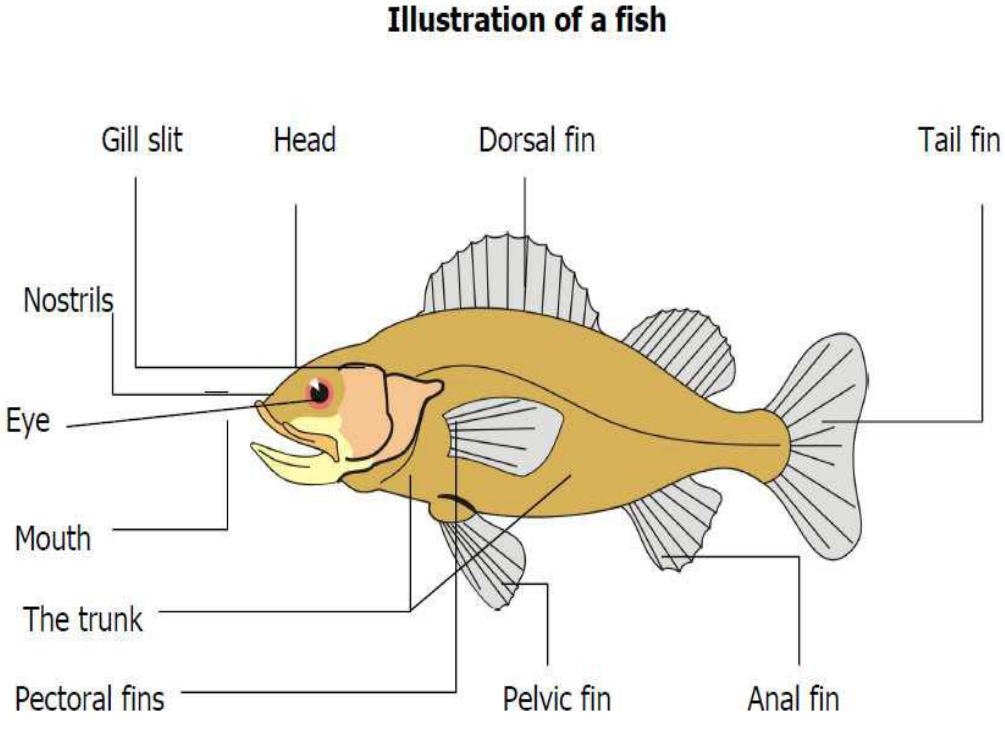
21. Monosex culture - This is the farming of only one sex (Male or Female) done by fish culturists through hormonal use, hand sexing or the use of supermales (YY) or superfemales (XX).

22. Polyculture or composite fish culture- It is a type of fish culture in which different types of fishes of different feeding habits are Cultured.

Points of Discussion

1. **What is a Fish ?**







Points of Discussion

1. **What is the difference between Fish and Fishes?**



Points of Discussion

1. What are the main sources of broodstock for hatcheries ?
2. Is it possible to successfully do aquaculture without utilizing wild resources?
3. Which one is better. A high FCR or Lower FCR?

Answers

1. **Wild (natural water bodies) and Hatchery reared**

2. **No. We need the wild for genetic diversity**

3. **A lower FCR is better**

(A lower FCR indicates better efficiency (utilisation) because it means the animal requires less feed to gain a certain amount of weight. High FCR values can signify inefficiencies in feed utilization, which can increase production costs and environmental impacts in aquaculture operations. Thus, aquaculture farmers aim to optimize FCR to improve profitability and sustainability)

Answers

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TAKE HOME QUESTION

Differentiate the following terms

- Aquaculture and fish farming
- Aquaculture and Agriculture

HISTORY OF AQUACULTURE

- Aquaculture is not new and according to Rabanal (1988), some form aquaculture existed for about 4000 years
- There are various theories of how aquaculture came to be are advanced such as **Oxbow, Catch-and Hold, Concentration, and Trap-and-Crop**



https://openknowledge.fao.org/handle/20.500.14283/ag158e

HISTORY OF AQUACULTURE

1.Oxbow

- This theory says that aquaculture started when people took advantage of natural oxbow lakes.
- These lakes form when rivers change course, leaving behind water bodies with lots of fish.
- People improved these lakes and started adding more fish, leading to early fish farming.

HISTORY OF AQUACULTURE

2. Catch-and Hold

- This idea is based on the fact that rich people wanted fresh fish all year.
- Workers would catch wild fish and keep them in artificial ponds near homes, gardens, or even moats.
- This made it easier to access fish when needed and Carp (*Cyprinus carpio*) is one type of fish that thrived in this system.

HISTORY OF AQUACULTURE

3.Concentration

- In places with heavy rains and seasonal floods, fish spread across the land when it was wet.
- When the water dried up, fish gathered in small ponds.
- People began managing these ponds by catching big fish while letting small ones grow and reproduce.
- Over time, they improved these ponds to hold more fish.

HISTORY OF AQUACULTURE

4. Trap-and-Crop

- This theory comes from coastal areas where tides bring water and fish into natural lagoons.
- People built traps to let fish in but not out, making harvesting easier.
- Later, they started letting the fish grow bigger before harvesting, creating an early form of fish farming.

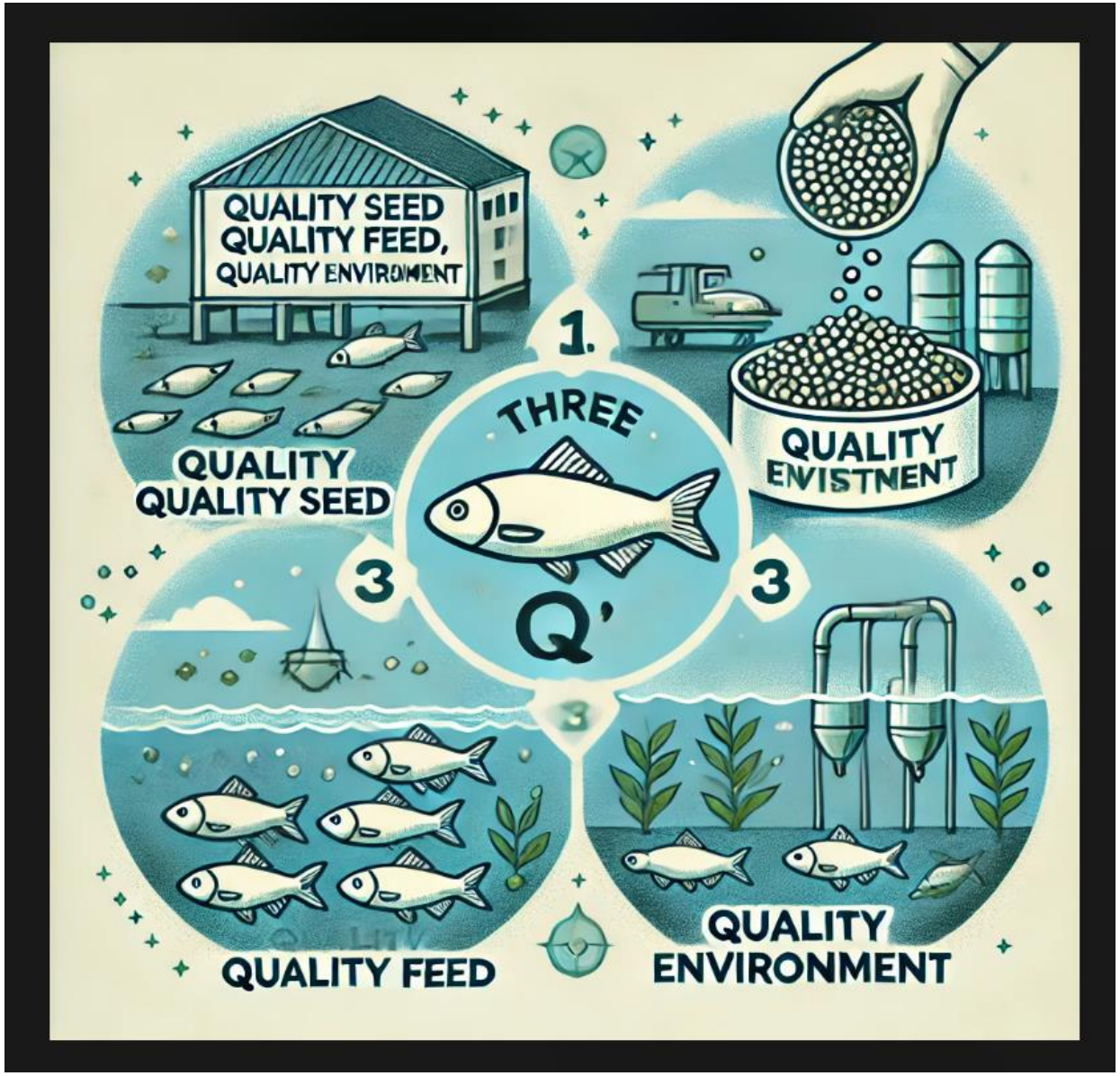
HISTORY OF AQUACULTURE

- Like all the culture practices, whether it is Agriculture, Horticulture and Animal Husbandry which includes, Poultry, Duckery, Dairy, Piggery and **Aquaculture too is based on three Q's**

What are these three Q's?

HISTORY OF AQUACULTURE

1) Quality seed (fingerlings)
2) Quality feed (Complete feed)
3) Quality environment (Water and culture facilities)



HISTORY OF AQUACULTURE

It is the responsibility of the workers of Extension services in the respective fields to make understand to the ignorant grass root level farmers about the immense utility and importance of the three Q’s.

HISTORY OF AQUACULTURE

QUESTION

Using four theories, discuss the history of aquaculture [20 Marks]

HISTORY OF AQUACULTURE

End